

CSCI 246 – Discrete Structures

Mike Wojnowicz

Fall 2026

Montana State University

Course Overview

Why Active Learning?

About Me

Discrete Math In Practice: Examples
From Machine Learning

Why Study Math In The Age of AI?

Student Survey

Room Change

Starting Friday 08/28, we will meet in Reid 201.

What is Discrete Math?

dis·crete / dis'krēt.

Adjective: Individually separate and distinct.

Synonyms: separate - detached - distinct - abstract.

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Warning: Details on the syllabus will be adjusted throughout the semester.

Syllabus Review

Let's review the [syllabus](#) !

How to read a math textbook

THE LEARNING CENTER • University of North Carolina at Chapel Hill



Tips for Reading Math Textbooks

Work through examples.

- Don't just skim (or worse, skip!) the examples. Instead, devote more of your focus to the examples. Working through them and verbalizing the main ideas behind them is a great way to test to make sure you're understanding what you're reading.
- Read with a pencil and paper in your hand and when you get to an example, work it out! Try not to look at the solution until you are done.
- When checking your work, make sure you understand each step and why.
- Practice the examples BEFORE you attempt homework problems and then try to do your homework without referring back to the example

Attending class is positively related to course performance

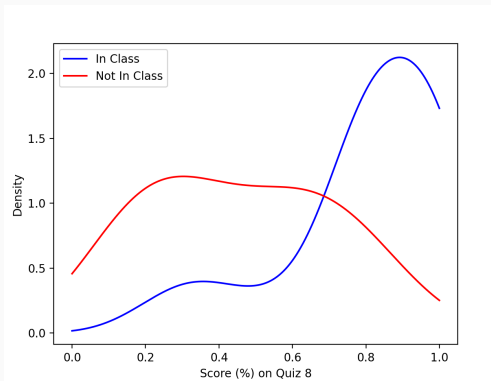


Figure 1: *Quiz Scores by Class Attendance.* Shown are scores on CSCI 246 Weekly Quiz #8 (Friday 10/24/25), broken in two groups, based on those who attended class on Monday 10/27/25.

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the worst way to teach is to talk.*

— **Paul Halmos** (1916–2006),
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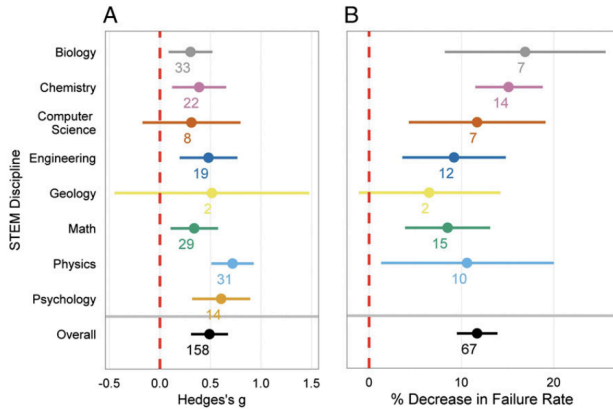
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True understanding comes from **engaging with problems actively**, which often requires a **shift in perspective** before a new solution becomes visible.

Why Active Learning?



Why Active Learning?



Freeman et al. (2014). *Proceedings of the National Academy of Sciences*.

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Michael Wojnowicz

Assistant Professor, GSoC @ MSU

Research Associate, Biostats @ Harvard

Postdoc, Machine Learning (CS) @ Tufts

Distinguished Data Scientist @ Cylance

(cybersecurity start-up, sold for \$1.4b)

PhD @ Cornell

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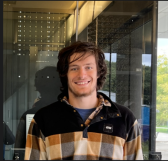
Research interests

- Probabilistic machine learning
- Scalable inference
- Modeling spatiotemporal data

MSU Probabilistic Machine Learning (ProbML) Lab

Lab Members

Jack Ruder (Ph.D. Student)



Diffusion models for time series

Kunal Das (Postdoc)



Spatiotemporal Causal Inference

Landry Clarke (Undergrad)



Deep Hierarchical Gaussian N-Grams

* Winner: MSU Undergraduate Scholars research award

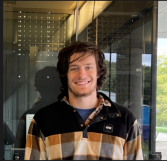
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My role: 30% teaching, 60% research, 10% service.

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Problem #1: Continuous Authentication



Continuous authentication:

continuously verify a user's identity
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Difficulty: Many 1000s of computer processes, sparsely observed.

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	Google Chrome
	Caffeine
	Activity Monitor
	Google Chrome Helper (GPU)
	Texifier
	Microsoft Outlook

Subgoal: Learn computer process start sequences which characterize the typical behavior of a user.

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Modeling Desiderata: probabilistic model with fast, reliable training

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The solution to this problem used almost every content area we're covering in this course

Discrete probability and counting

Complexity (Big-O notation)

Recursions

Graph theory

Proof techniques

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Can you explain why?

Problem #2: Early Cancer Detection

Healthy cell



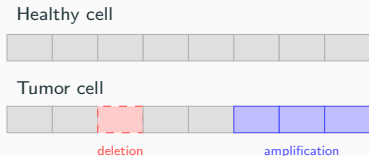
Tumor cell



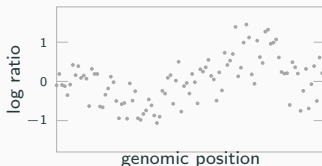
Somatic copy number

alterations (SCNAs): tumors gain and lose whole *segments* of DNA, carving the genome into contiguous blocks, each at its own copy number.

Problem #2: Early Cancer Detection

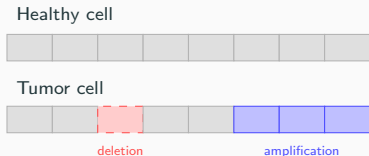


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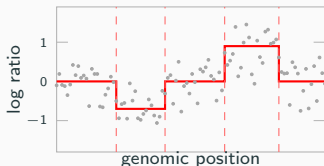
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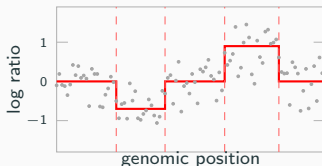
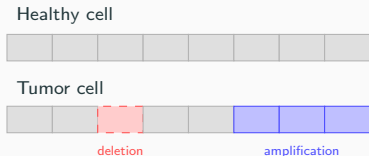
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Modeling Desiderata: probabilistic model that is accurate at low signal-to-noise, yet fast enough to scale

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Why study discrete math now?

Let's grant the strongest assumption

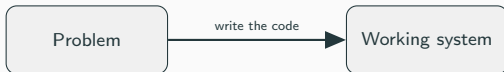
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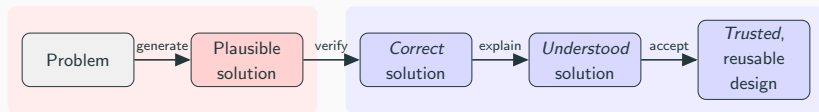


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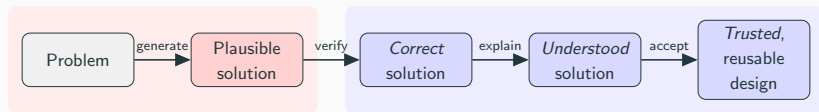
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...and this is the part that is still yours

Only the **first** stage was ever an explicit goal — and it is exactly the stage being automated. Every stage downstream of it is an act of **reasoning about correctness**. That is this course.

What that buys you: studies, life, work

Your studies

An AI can hand you correct homework. It cannot hand you the *ability* that the homework existed to build.

Induction, invariants, counting, graphs: these are the tools that let you say “**is right**” instead of “**looks right.**”

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Generative models are *ungrounded*: they optimize for output that *appears* satisfactory, not for the property it is supposed to indicate.

So you need a check that does not depend on the answer looking good. **One counterexample settles it.**

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Paid work becomes: *specify* precisely, *verify*, *explain*, and **take responsibility** for it.

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Goodhart's law: “When a measure becomes a target, it ceases to be a good measure.”
Passing the test was always the *measure*. Understanding was always the *target*.

Virtues of developing mathematical reasoning

It transfers to other subjects. Over one year, students taking post-compulsory mathematics improved at **conditional reasoning** more than matched literature students, despite having *no instruction in logic at all*.

Attridge & Inglis, *PLoS ONE* 8(7), 2013.

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It pays. For the marginal student, **one extra high-school math course** $\Rightarrow$ **$\sim 10\%$ higher earnings** via a shift into more cognitively skilled work.

Goodman, *J. Labor Economics* 37(4), 2019. Joensen & Nielsen, *JHR* 44(1), 2009: 20–25% per advanced course.

Don't like math? Tips for success

HOW TO THINK ABOUT IT



We start from scratch

No calculus needed. Nobody is behind on day one.

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Reach out early

I'm here to help!

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Introductory Survey

1. What is your name and major?
2. What is the most exciting thing about your major?
3. What do you hope will happen in this class?
4. What are some concerns you have about this class?
5. Anything else I should know?